

Name.....

Roll No.



ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI.

(An ISO 9001:2015 Certified College)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

B. TECH DEGREE FIRST INTERNAL TEST- NOVEMBER 2022

FIFTH SEMESTER – COMMON TO AD & CSE

CST 309 MANAGEMENT OF SOFTWARE SYSTEMS

Time: 2 Hours

Max.Marks: 50

PART A

Answer all Questions (3x5=15 marks)

Qn. No.	Questions	Marks	K level	CO & KL
1	Explain Spiral Model for software development with a neat diagram.	3	K1	CO1&K3
2	Introduce the concepts of Agile Development techniques and Agile Project Management.	3	K2	CO1&K3
3	What is Traceability Matrix (TM)?	3	K1	CO2&K3
4	List the characteristics of a good SRS document and their components.	3	K2	CO2&K3
5	Explain the terms (i) Reuse (ii) Configuration Management (iii) Host-target development	3	K1	CO3&K3

PART B

Answer five questions (7 x 5 = 35 Marks)

Module 1: Qn. No 6,7 and 8 – Answer any 2

Module 2: Qn. No 9,10 and 11 – Answer any 2

Module 3: Qn. No 12 is mandatory

Qn. No.	Questions	Marks	K Level	CO & KL
6	Differentiate between Waterfall model and Incremental model for software development.	7	K2	CO1 & K3
7	Describe the different process activities.	7	K1	CO1 & K3
8	a) List the principles of agile software development.	3	K2	CO1 & K3
	b) Briefly describe each of the four elements of the design model.	4	K2	CO2 & K3
9	a) If you are developing a library management software product, list the functional and non-functional requirements in it.	3	K3	CO2 & K3
	(b) Define the following: (i) Personas (ii) Scenarios (iii) User stories (iv) Feature identification	4	K1	CO2 & K3
10	Elucidate design concepts in software engineering.	7	K1	CO2 & K3

11✓	An Online Public Access Catalog (OPAC) is an <u>e-Library website</u> which is part of Integrated Library System (ILS), also known as a Library Management System (LMS), and managed by a library or group of libraries. Patrons of the library can <u>search the library catalog online to locate</u> various resources - books, <u>periodicals</u>, audio and visual materials, or other <u>items</u> under control of the library. Patrons may <u>reserve or renew</u> items, <u>provide feedback</u>, and <u>manage their account</u>. Draw the use case diagram for this project.	7	K3	CO2&K3
12✓	Compare white box testing and black box testing.	7	K2	CO3&K2

Name

Roll No.....



ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI.

(An ISO 9001:2015 Certified College)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

B. TECH DEGREE SECOND INTERNAL TEST- DECEMBER 2022

FIFTH SEMESTER – COMMON TO AD & CSE

CST 309 MANAGEMENT OF SOFTWARE SYSTEMS

Time: 2 Hours

Max.Marks: 50

PART A

Answer all Questions (3x5=15 marks)

Qn. No.	Questions	Marks	K level	CO & KL
1	Describe the Debugging approaches.	3	K1	CO3&K3
2	Illustrate Test driven development.	3	K2	CO3&K3
3	What is the importance of Software project planning?	3	K1	CO4&K3
4	Define software configuration management. List out different activities involved in configuration management.	3	K1	CO4&K3
5	Discuss the Software quality dilemma.	3	K2	CO5&K3

PART B

Answer five questions (7 x 5 = 35 Marks)

Module 3: Qn. No 6,7 and 8 – Answer any 2

Module 4: Qn. No 9,10 and 11 – Answer any 2

Module 5: Qn. No 12 is mandatory

Qn. No.	Questions	Marks	K Level	CO & KL
6	Explain the Software testing strategies.	7	K2	CO3 & K3
7	Describe the Formal and Informal Review techniques.	7	K2	CO3 & K3
8	What is Continuous Integration, Delivery, and Deployment ?	7	K1	CO3 & K3
9	Discuss in detail the COCOMO model of software project cost estimation technique.	7	K2	CO4 & K3
10	(a) List out and explain fundamental project management activities.	3	K2	CO4 & K3
	(b) Explain in detail about the Risk management in a software development.	4	K2	CO4 & K3

Name.....

Roll No.....

11	(a) Give a list of the factors that influence Software pricing.	3	K1	CO4&K3
	(b) Summarize the Kanban methodology and Lean Approaches.	4	K2	CO4&K2
12	Explain the following : CMMI and ISO 9001:2000.	7	K2	CO5&K3

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester B.Tech Degree Regular and Supplementary Examination December 2022 (2019 Scheme)

Course Code: CST 309**Course Name: MANAGEMENT OF SOFTWARE SYSTEMS**

Max. Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

Marks

- | | | |
|---|---|---|
| ① | Discuss the factors which are considered during the Components selection and design process. | 3 |
| ② | How does an agile approach help software developers to capture and define the user requirement effectively? | 3 |
| ③ | How do you prepare a software requirement specification? | 3 |
| ④ | Compare functional and non-functional requirements. | 3 |
| ⑤ | Differentiate between GPL and LGPL. | 3 |
| ⑥ | Compare White Box testing and Black box testing. | 3 |
| ⑦ | List out and explain the fundamental project management activities. | 3 |
| 8 | Discuss the role of using Backlogs and Sprints in SCRUM frameworks. | 3 |
| 9 | Explain cloud software characteristics. | 3 |
| ⑩ | Discuss software quality dilemma. | 3 |

PART B*(Answer one full question from each module, each question carries 14 marks)***Module -1**

- | | | |
|----|---|---|
| 11 | a) Design Boehm's Spiral model and its importance. | 7 |
| | b) Illustrate how the process differs in agile software development and traditional software development with a socially relevant case study. | 7 |
| 12 | a) Incremental model is better than water fall model for most business, e-commerce and personal systems. Justify the statement. | 7 |
| | b) Describe the relevance of using Pair programming and Refactoring during Agile development process. | 7 |

Module -2

- 13 a) Describe the various activities under Requirements engineering process. 7
 b) Outline the concept of traceability matrix and Requirements management planning. 7
- 14 a) What are Use cases? Draw the Use case diagram for an ATM. 7
 b) Explain Personas, Scenarios and Feature identification. 7

Module -3

- 15 a) What are design patterns? What are the essential elements of design patterns? 7
 b) Differentiate between Formal and Informal review techniques. 7
- 16 a) Differentiate between Top-down and Bottom-up Integration testing methods with suitable diagrams. 7
 b) Explain System testing and its variants. 7

Module -4

- 17 a) Explain plan driven development and project scheduling. 7
 b) Explain the Software Risk management process with the help of neat diagram. 7
- 18 a) What is algorithmic cost modelling? What problems does it suffer from when compared with other approaches to cost estimation? 7
 b) What is a critical path? Demonstrate its significance in a project schedule with the help of a sample project schedule. 7

Module -5

- 19 a) Compare CMMI and ISO 9001:2000. 7
 b) How is Software Quality achieved during Software engineering process? 7
- 20 a) Explain elements of Software Quality Assurance and SQA Tasks. 7
 b) Describe in detail about the Software Process Improvement (SPI) process. 7
